

MetalBin S.A.

Manufacturing of Industrial Metallic Containers Network Infrastructure Documentation and Firewall Policies

Technology and Infrastructure Department
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PART I — Company History and Context

1.1 Who We Are

MetalBin S.A. is a company founded in 2007 in the industrial zone of Alajuela, Costa Rica, specialized in the design, manufacturing, and distribution of metallic containers for solid waste management. Its products are present in municipalities, industrial complexes, shopping centers, and manufacturing plants throughout the Central American region. With more than 70 employees distributed across the production plant, logistics, sales, and administration, MetalBin operates in three continuous shifts manufacturing approximately 600 monthly units including 120L, 240L, 1100L containers and high-capacity industrial containers.

1.2 Organizational Structure.

Department	Responsibility	Critical Systems
Production / Plant	Control of cutting lines, welding, and automated painting	SCADA, PLCs, Win7 OT
Logistics	Delivery management, truck fleet, raw material inventory	ERP, GPS Fleet
Administration	Finance, HR, electronic invoicing to municipal clients	Windows 11, Office 365
IT / Infrastructure	Networks, servers, backups, internal technical support	Ubuntu Server, pfSense
General Management	Executive reports, strategic decision making	Zorin OS (executive laptop)
Maintenance	Printing of work orders and quality labels.	HP JetDirect Printer

1.3 History of the Technological Infrastructure.

When MetalBin started operations in 2007, the infrastructure was basic: computers running Windows XP, a shared printer, and an ADSL connection.

The accelerated growth between 2010 and 2014 forced the incorporation of industrial control systems (Siemens PLCs and proprietary SCADA) to automate sheet metal cutting lines and robotic welders.

In 2015, the first internal IT team was hired. The priority was to connect the PLCs to the corporate network to generate production reports in real time. This decision, made without a security evaluation, introduced the OT segment (Windows 7) directly into the corporate LAN network, a configuration that persists to this day.

In 2018, the HP LaserJet printer with a JetDirect interface was installed in the maintenance workshop.

To allow technicians to print from any device without complications, the IT team connected it directly to the host's physical network, outside of any firewall perimeter. Authentication was never configured. In 2021, pfSense was acquired as a firewall solution. The rules were configured permissively to 'not stop production', with allow-all policies in several segments and the promise to review them 'in the next quarter'.

That review never occurred. In 2024, a partial internal audit detected insufficient segmentation, uncontrolled communication paths between OT and the corporate network, and exposed services without authentication.

1.4 Documented Security Findings — May 2024 Audit

The audit report identified the following active issues:

- HP JetDirect printer (port 9100) exposed on the host network without authentication or encryption.
- Accepts print jobs from any device on the physical network. OT segment (Windows 7) with residual communication towards the corporate LAN through legacy rules that were never removed.
- Ubuntu Server in DMZ with SSH port (22) without source restriction — accepts connection attempts from any segment.
- No separation between the management network (pfSense WebUI) and the end-user network
- Firewall logs without active monitoring. Syslog server never configured. Default credentials on JetDirect printer — no password configured since installation.
- Windows 7 OT without security updates since 2020. Unpatched EternalBlue vulnerability (MS17-010) confirmed.
- Non-existent password policy. Several computers with passwords identical to the department name.

PART II — Network Topology

2.1 Network Segments

Segment	Network	Gateway	Devices
Corporate LAN	192.168.10.0/24	192.168.10.1	Win11 (.10), Zorin (.30), Win7-OT (.20)
DMZ — Server	192.168.30.0/24	192.168.30.1	Ubuntu Server (.10)
Host Network (Bridged)	192.168.1.0/24 *	192.168.1.1 *	Debian CUPS / JetDirect Printer
WAN / Internet	DHCP via NAT	—	pfSense em0 (VirtualBox NAT)

* The host network varies depending on the laboratory's physical router.

2.2 Assigned Fixed IPs

Machine	IP	RoI	Interfaz pfSense
pfSense (LAN)	192.168.10.1	Gateway LAN / Firewall	em1
pfSense (DMZ)	192.168.30.1	Gateway DMZ	em2
Windows 11	192.168.10.10	Corporate administration workstation	LAN — pfsense-lan
Windows 7 OT	192.168.10.20	Legacy industrial control (PLCs)	LAN — pfsense-lan
Zorin OS Lite	192.168.10.30	Management executive laptop	LAN — pfsense-lan
Ubuntu Server	192.168.30.10	File and SSH server,DMZ	DMZ — pfsense-dmz
Debian CUPS	192.168.10.222(DHCP)	JetDirect print server	Bridged — outside firewall

PART III — Firewall pfSense Rules

The following rules reflect the current configuration of the MetalBin S.A. network, including legacy policies and documented security gaps. They are configured in pfSense under Firewall > Rules.

3.1 WAN Interfaz (em0)

#	Action	Protocol	Source	Destination	Port	Description
1	BLOCK	Any	Any	Any	Any	Default deny all inbound (automatic pfSense)
2	PASS	TCP	Any	WAN Address	443	IT remote access

3.2 LAN Interfaz (em1) — 192.168.10.0/24

Rules are evaluated from top to bottom. The first matching rule wins. The order is critical.

#	Action	Source	Destination	Port	Description
1	BLOCK	192.168.10.20 (Win7)	192.168.30.0/24	Any	Win7-OT CANNOT reach the DMZ
2	BLOCK	192.168.10.20 (Win7)	!192.168.10.10	Any	Win7 blocked to everything except Win11
3	PASS	192.168.10.20 (Win7)	192.168.10.10	Any	Win7 ALLOWED to communicate with Win11 only
4	BLOCK	192.168.10.10 (Win11)	192.168.30.0/24	Any	Win11 CANNOT access the DMZ
5	PASS	192.168.10.30 (Zorin)	192.168.30.0/24	Any	Zorin ONLY one with access to the DMZ server
6	PASS	192.168.10.30 (Zorin)	Any	Any	Zorin full internet access
7	PASS	192.168.10.10 (Win11)	Any	Any	Win11 internet access
8	PASS	LAN subnets	pfSense 10.1	443	pfSense WebUI management from LAN
9	BLOCK	192.168.10.20 (Win7)	Any	Any	Catch-all — Win7 no internet access
10	PASS	LAN subnets	Any	Any	Default allow LAN — original legacy rule

Note — Rule 2 (Invert match in pfSense):

In the WebUI: Destination field → Single host → 192.168.10.10 → check 'Invert match' box. This means 'block to any destination EXCEPT Win11'.

3.3 DMZ Interfaz (em2) — 192.168.30.0/24

#	Action	Source	Destination	Port	Description
1	BLOCK	192.168.30.0/24	192.168.10.0/24	Any	DMZ cannot initiate connections towards LAN
2	BLOCK	192.168.30.0/24	192.168.10.1	443	DMZ cannot access pfSense WebUI
3	PASS	192.168.30.0/24	Any	80	Ubuntu outbound HTTP — apt updates
4	PASS	192.168.30.0/24	Any	443	Ubuntu outbound HTTPS
5	PASS	192.168.30.0/24	Any	53	Ubuntu DNS resolver
6	BLOCK	192.168.30.0/24	Any	Any	Default deny — everything else from DMZ

3.4 JetDirect Printer — Debian CUPS (Outside the Firewall)

The printer operates in Bridged mode with an IP on the host's physical network. pfSense DOES NOT control its traffic.

Current documented state:

- Port 9100 (JetDirect RAW) open without authentication — any device on the network sends print jobs.
- Port 631 (CUPS WebUI) accessible without password from any IP on the physical network.
- CUPS configured with 'Allow all' — no source restriction.
- JetDirect protocol without encryption — print jobs in clear text on the network.

Recommendations pending implementation:

- Move the printer behind the firewall into a dedicated segment 192.168.40.0/24.
- Restrict port 9100 only to authorized IPs.
- Enable authentication in the CUPS WebUI.

PARTE IV — Implementation in pfSense WebUI

Access from Windows 11 or Zorin OS at <https://192.168.10.1>

4.1 Create Aliases

Firewall > Aliases > Add — simplifies rules and facilitates future changes:

Alias	Type	Value	Description
Win11_Corp	Host	192.168.10.10	Corporate Windows 11 workstation
Win7_OT	Host	192.168.10.20	Windows 7 industrial control
Zorin_Exec	Host	192.168.10.30	Zorin OS executive laptop
Ubuntu_DMZ	Host	192.168.30.10	Ubuntu server in DMZ
LAN_Net	Network	192.168.10.0/24	Full LAN network
DMZ_Net	Network	192.168.30.0/24	Full DMZ network

4.2 LAN Rules— Step by Step

Firewall > Rules > LAN > Add (up arrow = insert at the top):

Rule 1 — Block Win7 towards DMZ

- Action: Block | Interface: LAN | Protocol: Any
- Source: Single host — 192.168.10.20
- Destination: Network — 192.168.30.0/24
- Description: Block Win7-OT to DMZ

Rule 2 — Block Win7 towards everything except Win11

- Action: Block | Source: 192.168.10.20
- Destination: Single host — 192.168.10.10 + marcar 'Invert match'
- Description: Block Win7 to anywhere except Win11

Rule 3 — Allow Win7 only towards Win11

- Action: Pass | Source: 192.168.10.20 | Destination: 192.168.10.10
- Description: Allow Win7-OT to Win11 only

Rule 4 — Block Win11 towards DMZ

- Action: Block | Source: 192.168.10.10 | Destination: 192.168.30.0/24
- Description: Block Win11 to DMZ

Rule 5 — Allow Zorin towards DMZ

- Action: Pass | Source: 192.168.10.30 | Destination: 192.168.30.0/24
- Description: Allow Zorin-Exec to DMZ — authorized only

Rules 6 and 7 — Internet for Zorin and Win11

- Pass | Source: 192.168.10.30 | Destination: Any — Zorin internet
- Pass | Source: 192.168.10.10 | Destination: Any — Win11 internet

Rule 8 — pfSense WebUI Access

- Action: Pass | Source: LAN subnets | Destination: This firewall | Port: 443
- Description: Allow LAN management access to pfSense WebUI

Rule 9 — Final Block Win7

- Action: Block | Source: 192.168.10.20 | Destination: Any
- Description: Final deny Win7 internet — catch-all

4.3 DMZ Rules — Step by Step

Firewall > Rules > DMZ > Add:

Rule 1 — Block DMZ towards LAN

- Action: Block | Source: DMZ subnets | Destination: 192.168.10.0/24
- Description: Block DMZ-initiated traffic to LAN

Rules 2-5 — Server internet outbound

- Pass | DMZ subnets | Any | Ports 80, 443, 53 — Updates and DNS

Rule 6 — Final Deny all DMZ

- Block | DMZ subnets | Any — Default deny all other DMZ traffic

4.4 Final Order of LAN Rules

Pos	Action	Source → Destination	Description
1	BLOCK	Win7 → DMZ	OT does not reach server
2	BLOCK	Win7 → !Win11	OT isolated from full network except Win11
3	PASS	Win7 → Win11	Only permitted communication for Win7
4	BLOCK	Win11 → DMZ	Corporate does not access the server
5	PASS	Zorin → DMZ	Only VM with access to the server
6	PASS	Zorin → Any	Executive laptop free internet
7	PASS	Win11 → Any	Corporate free internet
8	PASS	LAN → pfSense:443	Firewall WebUI management
9	BLOCK	Win7 → Any	Catch-all block OT internet
10	PASS	LAN → Any	Default inherited rule

PART V — Expected Connectivity Matrix

Source	Destination	Result	Rule
Win11 (10.10)	Internet	PASS	LAN #7
Win11 (10.10)	pfSense WebUI	PASS	LAN #8
Win11 (10.10)	Ubuntu DMZ (30.10)	BLOCK	LAN #4
Win11 (10.10)	Win7 OT (10.20)	PASS	LAN #10 default
Win11 (10.10)	Debian CUPS	PASS	Outside firewall — host network
Win7 OT (10.20)	Win11 (10.10)	PASS	LAN #3
Win7 OT (10.20)	Ubuntu DMZ	BLOCK	LAN #1
Win7 OT (10.20)	Internet	BLOCK	LAN #2 y #9
Win7 OT (10.20)	Debian CUPS	PASS	Outside firewall — host network
Zorin (10.30)	Ubuntu DMZ (30.10)	PASS	LAN #5 — only allowed one
Zorin (10.30)	Internet	PASS	LAN #6
Zorin (10.30)	pfSense WebUI	PASS	LAN #8
Zorin (10.30)	Debian CUPS	PASS	Outside firewall — host network
Ubuntu DMZ	LAN completa	BLOCK	DMZ #1
Ubuntu DMZ	Internet HTTP/S	PASS	DMZ #3 y #4
Debian CUPS	Cualquier red	PASS	No firewall — directly exposed

PART VI — Troubleshooting and Monitoring

6.1 pfSense WebUI Tools

- Diagnostics > Ping — Verify connectivity from pfSense to each segment
- Diagnostics > Packet Capture — Capture real-time traffic per interface
- Status > System Logs > Firewall — View blocked packets and the rule that acted
- Diagnostics > States — View all active connections in the firewall

6.2 Verification from the VMs

From Win11 (cmd):

```
ping 192.168.10.1      # Gateway pfSense LAN — must respond
ping 192.168.30.10    # Ubuntu Server — must FAIL from Win11
ping 8.8.8.8          # Internet — must work
```

From Win7 OT (cmd):

```
ping 192.168.10.10    # Win11 — must WORK
ping 192.168.30.10    # Ubuntu DMZ — must FAIL
ping 8.8.8.8          # Internet — must FAIL
```

From Zorin OS (terminal):

```
ping 192.168.30.10    # Ubuntu Server — must WORK (Zorin only)
ssh usuario@192.168.30.10 # SSH access to the server
```

From Ubuntu Server (terminal):

```
ping 192.168.30.1    # Gateway DMZ — works
ping 192.168.10.10    # Win11 — must FAIL
curl -I https://google.com # Internet — works
```

6.3 Boot Order

- pfSense — ALWAYS first. Without it, there is no DHCP or routing.
- Ubuntu Server — second, ready before Zorin connects.
- Windows 11, Windows 7 OT, Zorin OS — any order. Debian
- CUPS — independent, any time.

6.4 Recommended Snapshots

- pfSense — final rules applied and verified
- Win11 — static IP and base configuration
- Win7 OT — static IP configured
- Zorin OS — static IP and DMZ access verified
- Ubuntu Server — netplan applied, SSH active
- Debian CUPS — CUPS installed, JetDirect printer added